



Optimization of the EfficientNet-Swin Transformer Hybrid Model with Dynamic Attention and Multi-task Learning for Accurate Palm Fruit Classification

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Abstract: Manual grading of the maturity of oil palm fruits is subjective and unproductive in an industrial context. This paper presents a hybrid EfficientNet-Swin Transformer model with Dynamic Attention and Multi-Task Learning (MTL) for the effective classification of Fresh Fruit Bunches (FFB) maturity. The dataset employed in that was of 12.550 RGB images distributed within five classes: Defect, Overripe, Ripe, Underripe, and Unripe. The model integrates local feature extraction of EfficientNet-B3, global dependency modeling of Swin Transformer, and dynamic attention for adaptive feature refinement. Using 5-fold cross-validation during training, the model achieves an overall accuracy of 97.16% ($\pm 0.47\%$), precision of 97.17%, recall of 97.16%, F1-score of 97.16%, and mean AUC of 99.79%, higher than the baseline EfficientNet-B3 (95.21%). Defect detection-based multi-task learning scored an IoU of 0.7305 and a Dice coefficient of 0.7342, respectively. The model demonstrates good robustness against the variations in lighting conditions and visual diversity, thus proving to be an effective approach for online grading of palm oil fruits at plantations and factories.

Keywords: EfficientNet, Swin transformer, Dynamic attention, Multi-task learning, Palm fruit classification.

1. Introduction

As the world's largest producer of palm oil, Indonesia plays a major role in the global supply chain. Riau Province is always a region that owns the largest area of oil palm plantations in Indonesia, from 2.86 to 3.49 million ha or 19-21% of the Indonesian total plantation [1].

In the global palm oil industry, the quality of fresh fruit bunches (FFB) is important throughout the process. The ripeness of the fruit affects how much oil can be collected and the quality of that oil. It also changes how easy it is to refine the oil. However, current methods to check if fruit is ripe rely on people making judgments, which can be subjective, inconsistent, and time-consuming [2].

To address these challenges, AI-driven non-invasive or minimally invasive methods (including spectral sensing, hyperspectral imaging, and computer vision, among others) have become

transformative solutions. These methodologies have been backed by machine learning and deep learning, which can learn visual patterns such as color, texture, and chromatic features of FFB to classify the ripeness more objectively and reproducibly. Smartphone-based technologies extend these capabilities by providing low-cost, user-friendly, real-time field solutions [3].

Despite these improvements, even though there can be milestones with CNN and multispectral-based methods, major challenges remain and hinder their application in the real world. These challenges can be classified as environmental robustness (lighting condition changes, shadows, occlusions from leaves/other fruits, or perspective distortion) since the model performance often fails under these covering conditions [4].

To further clarify the hurdles, challenges can be divided into the following categories: Single-Task Methods: Some researchers examine only ripeness

classification rather than also learning from related tasks (e.g., defect detection), aiming to obtain more general models. **Data Constraints:** The small scale of data limits transferability to different cultivars and growing conditions. **Architecture Exploration:** There remains significant space to investigate the benefits of hybrid models that integrate local CNNs with global transformers; such combinations may leverage both detailed spatial features and broader contextual information, potentially improving classification performance for the visual complexity of FFBs [2].

This study was motivated by several issues related to classifying the maturity condition of oil palm fruit bunches (FFB) under current circumstances [5]. The human-based assessment system utilized so far is subjective as well as inefficient in the first place. Such reliance on human expertise is subjective, likely leading to irreproducibility of assessment and possibly causing crop loss; it cannot even fulfill the demands of industrial-scale plantation [6]. Furthermore, the autotelic-related inventions are of limited durability. However, the performance of these methods deteriorates sharply in practical field conditions, including changing illumination, occlusion by fruit parts, and background complexity [7].

Another issue is the restricted design of the mono-task model (e.g., classifying only one domain or only using a single data modality), which does not take the benefits of multi-task learning and multimodal data integration into account for better performance. What is more, sufficient model generalization is prevented by scarce training data. The lack of a large, diverse, and annotated FFB data pool impedes the development of models that are able to handle different palm oil species, regions, and farming traditions. Lastly, the advanced hybrid architectures have not been explored. The full potential of hybrid models, which can leverage the strengths of CNN for local feature extraction and Transformers for modeling global context, is not yet harnessed in this specific agricultural application.

Motivation to Develop a Hybrid Architecture of CNNs and Transformers It is recognized that the hybrid architecture of CNNs and transformers is superior to the single models in terms of embedding learning. This technique is considered a potential solution to the limitations of previous methods, utilizing effective dynamic attention mechanisms and enabling multi-task performance on devices with low computational power [8]. CNNs like ResNet50 have also shown outstanding progress with additional data augmentation and parameter fine-tuning, and methods based on object detection (like YOLOv4) could be used for a real-time comparison [9].

However, substantial challenges remain in complex field conditions, such as varying illumination, dense canopy structures, and incomplete modelling of task-specific dependencies [10].

Three major research gaps are identified: firstly, there has been little investigation of hybrid CNN-Transformer architectures that can combine the local feature extraction ability of a CNN and the global context understanding capability of transformers [11]. Second, this domain has not yet explored multitask learning (e.g., combining maturity classification with defect detection) to enhance data efficiency and model generalizability [12]. Third, most works remain based on a single sensor modality (RGB), underexploiting spectral information that may enhance discrimination at early stages [13]. Research under real field conditions with different cultivars, cultivation practices, and environments is also scarce.

We adopt a unified framework that combines the strong co-features from EfficientNet [14]. and the global context focusing of Swin Transformer and boosts them with a dynamic attention module to adapt features more responsively. This brief overview of the hybrid system aims to enhance robustness against changes in lighting and viewpoint. It utilizes a multi-task learning scheme to exploit the synergy between them for boosting data efficiency and model generalization. To alleviate the insufficiency of data, transfer learning and multimodality (RGB images with spectral information) are the main strategies [15]. The expected determinative criteria to separate maturation degrees are difficult to discriminate for age determination using multispectral imaging [13].

This study holds both theoretical and practical significance. In terms of its applications, the proposed system can be applied to enhance the accuracy and efficiency in the determination of maturity at both the plantation and factory levels, which influenced OER improvement and labor cost reduction as well as end product quality stabilization. Methodologically, the contribution lies in the powerful hybrid architecture design, the under-explored multitask learning scheme, and the extensive validation under real field conditions. The experimental methodology involves thorough performance evaluation on actual FFB datasets, generalization testing across diverse field scenarios, and real-time testing on embedded systems [16].

The structure of this paper is as follows. Section 2 provides an overview of the literature on CNN-based maturity classification, transfer learning, EfficientNet, Swin Transformer, multispectral imaging, and real-time grading. 3. Research Methodology This section presents the dataset, image preprocessing, initial hybrid architecture, multitask

learning, and evaluation methods. 4. Results and discussion. Accuracy, performance, and robustness. Comparison with the baseline. Component contribution analyses are discussed in Section 4 of this paper. Section 5, Conclusions and Future Work, contains the conclusion, practical implications, limitations, and future work.

2. Literature review

The classification of the oil palm fruit ripeness has been widely studied by using several network architectures, including convolutional neural networks (CNNs) and transformer-based models, and by implementing transfer learning and multitask learning methods that enhance the accuracy and robustness of the system.

Limitations of conventional techniques manual grading remains subjective and labor-intensive, producing inconsistent labels and operational bottlenecks in large estates [17]. Classical computer-vision pipelines-color thresholding (HSV/YCbCr), handcrafted texture descriptors (LBP/GLCM), and shallow classifiers (k-NN/SVM) - depend on careful, per-site calibration and are brittle under illumination changes, specular highlights, mud/dust artifacts, occlusions, and complex backgrounds; prior surveys have repeatedly noted limited cross-site generalization in plantation settings [18]. Although computationally light, these methods struggle precisely where real deployments demand stability: variable lighting, camera heterogeneity, and cultivar diversity [7].

For comparison, the work by [5] proposed a successful transfer learning method by modifying the existing implementations of EfficientNet. Their research applies the EffNetB0-LR and EffNetB4-LR models to the FFB for defect detection, using EfficientNet-B0, which has computational efficiency, and EfficientNet-B4, which allows for deeper features, combined with an LR classifier. These architectures present a state-of-the-art practice relying on pure CNNs and are well-suited for resource-limited computing environments. However, as CNN-based models, their long-range context modelling and global dependencies between fruit bunches are inherently less powerful. Our work intentionally pursues counters to these theoretical restrictions, for which we not only transfer learn a state-of-the-art CNN (EfficientNet-B3), but also include global context information modelling with Swin Transformer and Dynamic Attention techniques in our pipeline by considering feature fusion based on it adaptively.

Positioning relative to modern deep models. CNN-only classifiers deliver high accuracy yet primarily capture local texture/color cues and can under-represent long-range dependencies across fruitlets; transfer learning helps but can be sensitive to domain shift. EfficientDet-like detector backbones are latency-efficient but detection-focused, which can miss pixel-level maturity signals in densely packed bunches, with reported accuracy plateaus under field noise [19]. Transformer-only models (ViT/Swin) provide powerful global modeling but are more data-hungry and training-sensitive, especially in limited-labeled agricultural imagery. Our approach is positioned to bridge these gaps: a hybrid EfficientNet-Swin model with dynamic attention and multitask learning leverages EfficientNet's compute-efficient low-level stability while injecting hierarchical self-attention for multiscale context; multitask heads are chosen to encode physically meaningful ripeness correlates and are loss-weighted to avoid negative transfer. This combination targets field robustness and real-time feasibility, differentiating our method from conventional pipelines, CNN-only classifiers, detector-oriented designs, and transformer-only stacks.

Manual determination of ripeness in oil palm fruit has been a relatively subjective and laborious activity until now, leading to high interest in developing machine vision systems that offer automated and objective methods based on deep learning [20]. Other approaches tried are well-established methods of machine vision and machine learning (its main advantage being speed, as it always works the same). Nonetheless, Convolutional Neural Networks (CNNs) are currently the state of the art and the preferred method for this purpose [21]. Models like ResNet-50, known for its 97% accuracy in classification tasks, demonstrate their effectiveness [22]. The field has been further advanced through transfer learning, where applying pre-trained weights to related datasets consistently boosts performance [23].

More recent models that have been adopted are architecturally complex in order to optimize operational efficiency. The EfficientDet-Lite2 model, for example, makes a trade-off that is more appropriate to the relatively lower throughput of that network type, leading to about 84% accuracy. It consists of an efficient feature extraction backbone, EfficientNet, and a novel erasing mechanism to suppress background features [24].

A paradigm change is being witnessed with the advent of Vision Transformers (ViT) and hierarchical versions such as Swin Transformer. The results suggest that transformers have more capacity to

model long-range dependencies and fine-grained color, texture, and shape variations than conventional CNNs in some types of fine-grained low-to-mid-level object classification tasks [25]. This sensitivity to turgor presents them as highly specialized, for example, in determining the ripeness of oil palm fruit.

Building on the complementary strengths of both these methods, hybrid CNN-Transformer architectures are emerging as the current frontier [26]. The emerged EfficientNet-Swin Transformer (ETET) hybrid model, which we use ET as short for, is a straightforward and reasonable continuation of the way of thinking. This architecture deliberately uses EfficientNet as a very efficient backbone for extracting the initial features, benefiting from its known computational cost and efficiency. This model complements the Swin Transformer block by serving as a multiscale context extractor. It captures global context information and complex patterns that are inaccessible to pure CNNs, driving classification accuracy even further beyond current benchmarks without inducing excessively heavy computation for real-world deployment.

To further improve the stability, we introduce a dynamic attention mechanism into our model. This enables the system to learn and concentrate on the most contextually relevant parts of the fruit, and it improves performance under challenging environmental conditions such as those imposed by changing light levels or different viewing points. Note also that the architecture applies multi-task learning. Co-training the model to predict not only the primary ripeness label but also auxiliary properties—such as fruit size, dominant color, or a

maturity index—forces the system to build more generalized and robust feature representations. This generalization strategy, which has been shown to work well in various fine-grained recognition tasks, enables better cross-dataset and environmental generalization [8].

In brief, we transition from using a single strong network like ResNet-50 or EfficientNet to a smart and hybrid system. The Another Step model (ETET) is a particular realization of this next generation with strong local feature extraction and robust global context understanding, and it was trained not only for ripeness classification but also for resolution refinement (multitasking-based reinforcement learning), targeting attaining a unique state-of-the-art in accuracy, efficient computation, and robustness.

3. Methodology

3.1 Proposed model

Fig. 1. In this paper, a novel hybrid deep learning framework is proposed for palm oil fruit classification and segmentation by integrating the EfficientNet-B3 model, Swin Transformer, and dynamic attention mechanisms. The pipeline starts with image processing such as normalization [0,1] and augmentation (rotation, shift, flip, zoom).

EfficientNet-B3 is used as the feature extraction backbone (7 MBConv blocks with depthwise separable convolution in 4-5 layers) based on ReLU6 activation. All blocks generate feature maps with different spatial sizes and channels through squeeze-excitation modules.

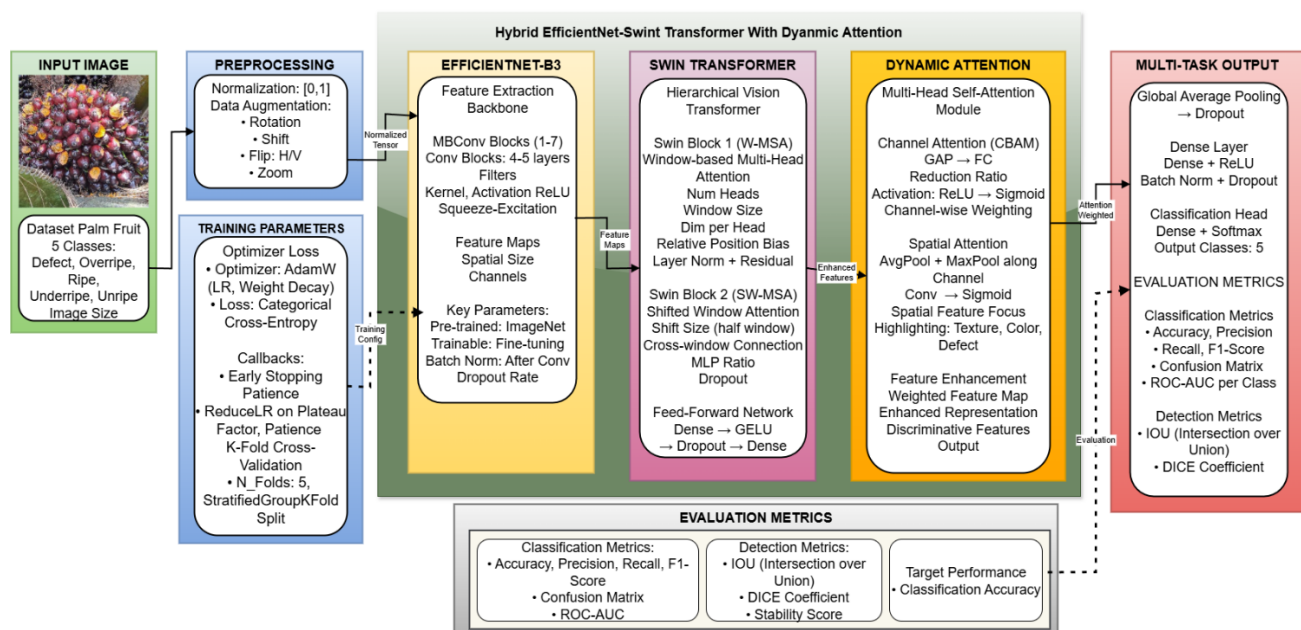


Figure. 1 Proposed Model

These characteristics are input into the Swin Transformer with two blocks: the Swin Block 1 is equipped with W-MSA with relative position bias and layer normalization, while the Swin Block 2 uses SW-MSA connected across windows to facilitate efficient long-range dependency modeling.

Feature promotion is achieved by multi-head self-attention followed by CBAM channel attention. It uses global average pooling and fully connected layers with a reduction ratio for channel-wise weighting. Spatial attention: The spatial attention achieves distinguishing emphasis on texture, color, and defects through average pooling and max pooling followed by convolution and the sigmoid activation in order to derive discriminative attention-weighted feature maps. It is capable of multi-tasking and has a final output head where the structural representation, with global average pooling, dropout, and dense layers with batch normalization, is concatenated to a 5-class softmax classification for Defect, Overripe, Ripe, Underripe, and Unripe.

Training uses the AdamW optimizer with a learning rate schedule & weight decay and optimizes categorical cross-entropy loss. Regularization involves the use of early stopping and ReduceLROnPlateau. Stratified K-fold cross-validation is used to maintain class distribution.

The model robustness is tested by the classification metrics (i.e., accuracy, precision, recall, F1-score, confusion matrix, and ROC-AUC) and detection metrics (e.g., IoU/overlapping ratio measure method, DICE coefficient measure formula, and stability score) to thoroughly evaluate the model performance in red palm oil fruit ripeness classification.

3.2 Dataset

For this study, we examined a set of fresh palm oil fruit bunches (FFB), as outlined in Table 1, which had been subdivided into five maturity classes: overripe, ripe, underripe, unripe, and defective. For

this study, oil palm fruit classification was performed into five classes based on maturity grade and morphological characteristics. Overripe fruit has an increased FFA transmission factor and is over-darkened, turning dark to red, purple, or a dark color, with the surface color lost. Ripe fruit has a greater size than unripe fruit, and the surface color is dark red, with a high transmission coefficient and many seeds compared to overripe fruit. Underripe fruit contains at least ten empty fruit sockets and has higher concentrations of flavonols and anthocyanins compared to ripe and overripe fruit. In contrast, unripe fruit has low output due to being dry (pistillate) and is characterized by dark purple or black skin with no empty sockets. Worst of all, defective fruit is classified as dry and shriveled, rotten and moldy, or bruised. This indicates that unripe oil palm fruit exhibits abnormal conditions and signs of dryness.

According to Table 2, a comparison of classification and segmentation datasets, we use two independent but correlated datasets for classification and segmentation/defect detection tasks, which are the palm oil fruits. The classification dataset includes 12,550 images covering the classes of Defect, Overripe, Ripe, Underripe, and Unripe, split into a training set (80.4%, 10,085 images), a validation set (13.0%: 1,630 images), and a test set (6.6%: 835 images). On the other hand, the dataset with respect to segmentation/defect contains 12,450 images designed for partitioning into 80.3% training data (9,986 images), 12.9% validation data (1,608 images), and 6.7% test data (856 images), which only focuses on four types of fruit defects: crack, rotten, abnormal color, and damage. A major contrast between these two datasets occurs in the annotation process: the segmentation/defect dataset is all labeled and elaborately masked for different defect types, while for the classification dataset, only the class label, but no segmentation mask-producing step, is needed since it only needs to classify fruit ripeness level as a whole.

Table 1. Class of palm fruit

No	Class	Morphological Characteristics
1	Overripe	Increased FFA transmission factor, loss of surface colour: turns dark red, purple, or dark coloration; reduced level of flavonoids and anthocyanins.
2	Ripe	Larger than unripe, Surface color: changes to dark red, High transmission coefficient, Seed: more than that in overripe
3	Underripe	At least ten empty fruit sockets, Higher content of flavonoid and anthocyanin when compared to ripe and overripe
4	Unripe	Low output due to dry fruit (Pistillate), Dark purple or black colored fruit skin, no empty sockets in the fruits.
5	Defect	Dry and deflated, rotten and moldy, to bruised

Table 2. classification and segmentation/defect dataset

Task	Split	Images	Percentage %	Classes/Types
Classification	Train	10.085	80.4%	Defect,Overripe,Ripe,Underripe, Unripe
	Valid	1.630	13.0%	Defect,Overripe,Ripe,Underripe, Unripe
	Test	835	6.6%	Defect,Overripe,Ripe,Underripe, Unripe
	Total	12.550	100%	5 classes
Segmentation/Defect	Train	9.986	80.3%	Crack, Rotten, Abnormal Color, Damage
	Validation	1.608	12.9%	Crack, Rotten, Abnormal Color, Damage
	Test	856	6.7%	Crack, Rotten, Abnormal Color, Damage
	Total	12.450	100%	4 defect types

The segmentation mask generation process was performed by computer vision image processing techniques, namely edge detection, color segmentation, and intensity analysis. Masks are generated for crack defects with masks created by Contrast Limited Adaptive Histogram Equalization (CLAHE) preprocessing (contrast optimization, window size 32) and Canny edge detection ($\sigma=30-100$), following morphological filtering using elongated kernels (1×15 , 15×1) to identify the crack pattern of at least length >20 pixels. Rotten and abnormal color defects are detected via HSV color space transformation, respectively, by multi-range thresholding and morphological closing (5×5 kernel) for region consistency, while damage defects adopt adaptive thresholding for dark areas and edge detection applying Gaussian smoothing ($\sigma=5$). All found masks are further processed by a set of post-processing steps: the application of the polygon approximation (vectorization) approach using the Douglas-Peucker algorithm with an ϵ value equal to 1% of the perimeter and zero-base contour filtering according to minimal area as follows: Crack $\geq 50\text{px}^2$, Rotten $\geq 100\text{px}^2$, Abnormal Color $\geq 150\text{px}^2$, and Damage $\geq 80\text{px}^2$. This annotation step makes it possible for each image out of the 12,450 in the encoding/defect dataset to be correctly labeled and masked based on the types of defect found by first localizing the fruit ROI (Region of Interest) using HSV-based segmentation, followed by morphological operation, before proceeding with the defect annotation process.

The classification and segmentation datasets are publicly available at: <https://datasetpalmoil.com/>

The classification dataset is divided into three subfolders: train, test, and valid, with images of maturity classes. On the other hand, the segmentation dataset consists of three main folders: defect_annotations_train, defect_annotation_test, and defect_annotation_val. There are three subfolders in each segmentation folder, i.e., labels (annotations for defect types), masks (binary or segmentation images of defects), and visualizations

(visual overlays of annotated defects on original images). It is this structured framework that allows for efficient data management and seamless deep learning integration in the classification as well as segmentation task. In addition to using internal datasets for training, we also use external datasets from Kaggle at the URL: <https://www.kaggle.com/datasets/advalhakim/oil-palm-class>.

For these computationally intensive tasks, all simulations, model trainings, and optimizations were performed on a high-performance workstation with an Intel Octa Core i9-14900HX (8x 3.30 GHz bis 5.80 GHz), 32 GB DDR RAM, and a NVIDIA GeForce RTX 4060 8 GB GDDR6 VRAM graphics card. This hardware was purposefully selected for maximizing parallel computing on the GPU. It can handle large and memory-consuming, complex data sets and models, which makes the development of this research feasible.

3.3 Data preparation

The preprocessing step scales the pixel value into the $[0, 1]$ range to facilitate training and speed up convergence. Data augmentation is performed in real-time, including random rotation, horizontal/vertical shift, flip (horizontal and vertical), and zoom, to enhance dataset diversity and prevent overfitting. Input image size is standardized to 224×224 pixels according to the EfficientNet-B3 and Swin Transformer architecture standards.

Training Settings The training parameters are based on AdamW with a learning rate schedule and weight decay for model weight regularization. This is a multi-class classification of 5 classes (palm oil fruit ripeness), and the loss function used in this experiment is the categorical cross-entropy. Callbacks have been implemented as early stopping with patience such that training can be halted if validation loss is not improving, and ReduceLROnPlateau, which reduces the learning rate adaptively when a plateau is noted to tune convergence.

The validation approach is through N-K folds cross-validation and Stratifiedgroupkfold split such that all folds contain approximately the same proportion of each class. This method gives a more stable performance estimation of the model that avoids bias in testing and results in good generalizations based on unseen data for credible and reliable oil palm fruit ripeness detection.

3.4 Hybrid efficientNet-swin transformer

The Integrated Hybrid EfficientNet-Swin Transformer is designed to leverage the benefits of both vision transformer models and convolutional neural networks for end-to-end feature learning in palm oil fruit image analysis.

By combining the Swin Transformers' capability of long-range dependency modelling and global context with the EfficientNet-B3's capability of detailed local texture feature extraction, including color, edge, and texture pattern, this hybrid model is the perfect way to tackle both local and global features.

3.4.1. Swin transformer pathway

Input and patch embedding (a common implementation uses Conv 4×4, stride 4):

$$\mathcal{Z}^{[3]}(0) = \text{Conv}_{4 \times 4, s=4}(X), \mathcal{Z}^{[3]}(0) \in \mathbb{R}^{(H/4) \times (W/4) \times C_0} \quad (1)$$

Instead of flattening the patch and projecting it linearly, many Swin implementations use 4×4 convolution with a stride of 4 to reduce the resolution while increasing the channel to C_0 ; The result is equivalent to linear embedding on the patch. Number of tokens (patch-flatten equivalent) and linear embedding alternative:

$$N = (HW)/p^2, p = 4; \mathcal{Z}^{[3]}(0) = PE + b, E \in \mathbb{R}^{(3p^2) \times C_0} \quad (2)$$

Conceptually, cutting a $p \times p$ patch produces N tokens. The classic alternative is to flatten each patch and then project it by E and bias b in a single $M \times M$ window (head k):

$$q = XW_Q^{(k)}, k = XW_K^{(k)}, v = XW_V^{(k)} \quad (3)$$

where window features are mapped to query-key-value space via matrices $W_Q^{(k)}$, $W_K^{(k)}$, and $W_V^{(k)}$ to measure similarity and combine high-value

information. Attention with relative position bias:

$$\text{Attn}^{(k)}(X) = \text{Softmax}\left((qk^T)/\sqrt{d_k} + \text{Brel}\right)v \quad (4)$$

where the score $(qk^T)/\sqrt{d_k}$ normalizes the scale of the header dimension d_k ; the sum Brel injects relative proximity between tokens. Softmax produces attention weights, which are then applied to a linear combination. v .

Multi-head self-attention and projection:

$$\text{MSA}(X) = \text{Concat}_{k=1}^h \left(\text{Attn}^{(k)}(X) \right) W_O \quad (5)$$

Pattern diversity is achieved by concatenating the h heads feature on the channel and projecting it back by W_O .

$$g = \text{GAP}\left(\mathcal{Z}^{[3]}(L)\right) \in \mathbb{R}^C \quad (6)$$

Here, the Global Average Pooling aggregates the channel responses to a C -sized vector g . At stands for global context.

3.4.2. Efficient pathway

Feature extraction through the EfficientNet-B3 architecture: Compound scaling (depth, channel width, resolution) with computational cost constraints:

$$d = \alpha^\phi d_0, w = \beta^\phi w_0, r = \gamma^\phi r_0, \alpha\beta^2\gamma^2 \approx 2 \quad (7)$$

where scale ϕ increases depth d , channel width w , and resolution r simultaneously; coefficients (α, β, γ) are selected so that an increase in $\phi \approx$ doubles the estimated FLOPs ($\approx \alpha\beta^2\gamma^2$). MBConv block scheme (summary): expansion $\rightarrow$ depthwise $\rightarrow$ SE $\rightarrow$ projection (+ skip if applicable):

$$U = \text{DWConv}\left(\delta\left(\text{Conv}_{1 \times 1}^{\text{exp}}(X)\right)\right) \quad (8)$$

where feature X is expanded through 1×1 convolution (ratio t), activated δ (e.g., Swish/ReLU), then depthwise conv is applied to capture local context per channel, resulting in U . Squeeze-and-Excitation (SE) integrated into MBConv:

$$z = \text{GAP}(U) \in \mathbb{R}^C \quad (9)$$

where channel vector z is obtained by spatially

averaging.

$$s = \sigma(W_2 \delta(W_1 z)) \quad (10)$$

where two full layers (bottleneck) transform z into channel amplification coefficients s in $[0,1] \mathbb{R}^C$ through activation δ and sigmoid σ . The U feature is scaled per-channel by s (Hadamard multiplication) to emphasize informative channels and suppress less relevant ones.

$$\mathcal{E}_s^{[3]} = f_s(X), s = 1, \dots, S, \mathcal{E}_s^{[3]} \in \mathbb{R}^{H_s \times W_s \times C_s} \quad (11)$$

where each stage produces a tensor $\mathcal{E}_s^{[3]}$. At different resolutions and channels, the initial level captures texture, and the final level captures semantics. Local Texture Features: These features include color details, edges, and texture patterns that are critical for defect and maturity identification.

3.4.3. Cross-feature fusion layer

Fusion aims to integrate global knowledge from Swin and local facts from EfficientNet in a fair comparison manner, such that they can be complementary to each other. This layer combines features from both pathways to overcome the limitations of a single model: Concatenation after channel/resolution alignment:

$$\mathcal{F}_s^{[3]} = \sigma \left(BN \left(W_s * \left[\widetilde{\mathcal{G}}_s^{[3]} \parallel \widetilde{\mathcal{E}}_s^{[3]} \right] \right) \right) \quad (12)$$

where Swin features $\widetilde{\mathcal{G}}_s^{[3]}$ and EfficientNet features $\mathcal{E}_s^{[3]}$ are combined, convolved by W_s , BN normalized, then activated by σ to produce $\mathcal{F}_s^{[3]}$.

Cross-scale aggregation with normalized weights:

$$\mathcal{F}^{[3]} = \sum_{s=1}^S \eta_s \mathcal{F}_s^{[3]}, \eta_s \geq 0, \sum_{s=1}^S \eta_s = 1 \quad (13)$$

where features of various scales are combined as a convex mixture so that the most informative scales dominate the final representation. Projection to class space and softmax:

$$\hat{y} = \text{softmax}(W_c \text{GAP}(\mathcal{F}^{[3]}) + b_c) \quad (14)$$

where $\mathcal{F}^{[3]}$ is summarized via GAP and projected onto class logits, then softmaxed into probabilities.

3.4.4. Fused multi-scale features

The Fused Multi-Scale Features module is an essential ingredient in the EfficientNet-Swin Transformer hybrid architecture to enable the synergy between the feature representations produced by the two pathways. This fusion is not shown in the diagram; instead, it is carried out implicitly via operations to utilise the complementary strengths of the two architectures. Fused Multi-Scale Feature Mathematical Representation. The operation is organized as follows: Multi-scale feature fusion is achieved by a combination of linear transformation and attention operation:

$$F_{fused} = \alpha \cdot \varphi(F_{swin}) + (1 - \alpha) \cdot \psi(F_{eff}) \quad (15)$$

F_{swin} These are the feature maps of the Swin Transformer pathway. F_{eff} These are feature maps from the efficient features pathway. φ and ψ are 1×1 convolutional projections matching the dimensions: α It is a context-dependent weighted parameter that automatically learns with the input content. The swin path builds a feature hierarchy through window-based attention and patch merging, ending with a $\vec{g}$ summary.

$$X \rightarrow \text{EfficientNetB3}(\text{MBConv} + \text{SE}) \rightarrow \mathcal{E}_s^{[3]}_{s=1}^S \quad (16)$$

The EfficientNet path generates multi-scale features using MBConv in conjunction with SE.

$$\begin{aligned} &(\mathcal{G}_s^{[3]}, \mathcal{E}_s^{[3]}) \rightarrow \text{Cross-FeatureFusion} \rightarrow \\ &\mathcal{F}_s^{[3]}_{s=1}^S \rightarrow \sum \eta_s \mathcal{F}_s^{[3]} \rightarrow \mathcal{F}^{[3]} \rightarrow \text{Head} \rightarrow \hat{y} \end{aligned} \quad (17)$$

Both paths are fused per scale and then aggregated for the final prediction.

3.5 Adaptive dynamic attention

Image Content Analysis: The purpose of this stage is to extract stable and informative content indicators to regulate adaptive attention. Indicators are calculated in the luminance and color domains, then summarized into robust global statistics. Luminance is calculated from linearized RGB channels ($\text{sRGB} \rightarrow \text{linear}$). The coefficients reflect the spectral sensitivity of human vision. To ensure that the indicators are not biased towards the lighting scale, the statistics are normalized (z-score)

Before being used in the downstream stage.

$$\tilde{z} = (z - \mu_z) / \sigma_z \quad (18)$$

Table 3. Notation list

Symbol	Description
$Z^{[3]}(0)$	Initial patch embeddings from input
$Conv_{4 \times 4, s=4}(X)$	4×4 convolution with stride 4 applied to input X
H, W	Height and width of input image
C_0	Initial channel dimension
N	Number of tokens (patches)
p	Patch size
PE	Positional encoding
b	Bias term
$E \in R^{(3p^2) \times C_0}$	Linear embedding matrix
q, k, v	Query, key, and value vectors
X	Input feature tensor
$W_Q^{(k)}, W_K^{(k)}, W_V^{(k)}$	Query, key, and value projection matrices for head k
$Attn^{(k)}(X)$	Attention output for head k
d_k	Dimension of key vectors
B_{rel}	Relative position bias
$MSA(X)$	Multi-head self-attention output
h	Number of attention heads
W_O	Output projection matrix
g	Global context vector from Global Average Pooling
$GAP(\cdot)$	Global Average Pooling operation
$Z^{[3]}(L)$	Feature tensor at layer L
C	Channel dimension
$\$d, w, r\$$	Depth, width, and resolution scaling factors
α, β, γ	Compound scaling coefficients
φ	Scaling parameter
d_0, w_0, r_0	Base depth, width, and resolution
U	Intermediate feature after depthwise convolution
$DWConv(\cdot)$	Depthwise convolution
$\delta(\cdot)$	Activation function (e.g., Swish, ReLU)
$Conv_{1 \times 1}^{exp}(\cdot)$	1×1 expansion convolution
t	Expansion ratio
z	Channel vector from spatial averaging
s	Channel attention coefficients
$\sigma(\cdot)$	Sigmoid activation function
W_1, W_2	Squeeze-and-Excitation network weights
$\mathcal{E}_s^{[3]}$	EfficientNet features at stage s
$f_s(\cdot)$	Feature extraction function at stage s
S	Number of stages
H_s, W_s, C_s	Height, width, and channels at stage s
$\mathcal{F}_s^{[3]}$	Fused features at stage s
$\mathcal{G}_s^{[3]}$	Aligned Swin Transformer features at stage s
$\widetilde{\mathcal{E}}_s^{[3]}$	Aligned EfficientNet features at stage s

W_s	Convolution weights for fusion at stage s
$BN(\cdot)$	Batch normalization
$\mathcal{F}^{[3]}$	Final aggregated features
η_s	Normalized aggregation weights
$\hat{y}$	Predicted class probabilities
W_c	Classification head weights
b_c	Classification bias
F_{fused}	Fused multi-scale features
F_{swin}	Swin Transformer feature maps
F_{eff}	EfficientNet feature maps
φ, ψ	1×1 convolutional projections
α	Context-dependent weighting parameter
$\tilde{z}$	Normalized content indicator
μ_z, σ_z	Mean and standard deviation for normalization
$S(x)$	Saliency map at pixel x
w_1, w_2, w_3	Saliency weighting coefficients
$\nabla Y(x)$	Gradient of luminance at pixel x
$Y(x)$	Luminance value at pixel x
μ_Y	Mean luminance
$F_{bc}(x)$	Color feature at pixel x
$F_{m\mu_c}$	Mean color feature
p	Quantile threshold for critical region
z	Content indicator vector
γ	Gate coefficient
F_g	Gate network weights
α_s, α_c	Dynamic weights for spatial and channel attention
F_s, F_c	Spatial and channel attention features
F_{mix}	Mixed attention features
M_c	Critical area mask
$W(x)$	Weighting function at pixel x
β	Critical region enhancement parameter
$F_{ref}(x)$	Refined features at pixel x

3.5.1. Critical region identification

Saliency $S(x)$ combines local contrast, edges, and color deviation so that defects/color variations stand out.

$$S(x) = w_1 |\nabla Y(x)| + w_2 |Y(x) - \mu_Y| + w_3 |F_{bc}(x) - F_{m\mu_c}|_2 \quad (19)$$

The critical area mask is created with an adaptive threshold; a stable practice is to use quantile p so that the proportion of the selected area is controlled.

3.5.2. Gate network

Content indicators are summarized into a vector z containing diversity, contrast, occlusion potential, and average saliency. This vector is mapped to gate coefficients $\gamma \in [0,1]$ that modulate the strength of the following attention module.

$$\gamma = \sigma(Fw_g^T z) \quad (20)$$

3.5.3. Dynamic weight calculation

Two scores are formed for each attention path, spatial, and channel, and then normalized with a softmax of temperature γ . When γ is high (content is “difficult”), the weight distribution becomes sharper, forcing the selection of more relevant modules; when γ is low, the weights become flatter.

3.6 Feature refinement & weighting

The two modules are combined linearly with dynamic weights (α_s, α_c) . Next, the detected critical area is enhanced by a mask M_c .

$$F_{mix} = \alpha_s F_s + \alpha_c F_c \quad (21)$$

$$W(x) = 1 + \beta M_c(x) \quad (22)$$

$$F_{ref}(x) = W(x) F_{mix}(x) \quad (23)$$

For $\beta > 0$, the contribution at critical locations increases additively-multiplicatively ($\text{gain} = 1 + \beta$), while non-critical locations remain unchanged ($\text{gain} = 1$). Optimally, this scheme increases the gradient in critical areas so that parameter updates are more guided by substantial evidence.

3.7 Multi-task learning with task-specific gates

It conducts joint classification and segmentation/detection by learning in a common framework. The Shared Encoder utilizes a large number of generic features, at no effective cost to the parameters. This allows for few-shot training on semantics while benefiting multiple tasks at once, all thanks to a transferable representation that stabilizes cross-task learning. A task-specific gating mechanism dynamically redistributes channel capacity from the encoder to each task and effectively suppresses negative interference, as well as promotes channel adaptation. The main task is aimed at dealing with the ripeness classification and aligns features to class ripeness probabilities such that accuracy is maximized for shared representations. The defect

segmentation/detection task in the auxiliary task is detecting and locating defect pixels/regions as an additional learning signal to boost the relationship of shared features without compromising classification accuracy.

3.8 Evaluation & analysis

Joint evaluation scores the model on both tasks simultaneously. The overall performance dynamics are measured by accuracy and loss, and the Classification Reports (Precision, Recall, F1-score) display the strengths, weaknesses, and biases specific to each class in the classification process. ROC curves and AUC measure classification discriminability independent of thresholds, while confidence distribution and calibration analysis ensure reliable probability scores. Performance evaluation in segmentation/detection problems is done by means of IoU (Intersection over Union) and the Dice coefficient, which measure the pixel-wise accuracy in the defect area localization, as well as the spatial overlap between predicted and ground-truth masks, guaranteeing a robust presence of defect detection, in addition to the classification accuracy.

4. Result

Five maturity stages of fruit were classified: Defect, Overripe, Ripe, Underripe, and Unripe. The input images were resized to 224×224 pixels as required by most pretrained models. The k value of 5 was applied to perform a 5-fold cross-validation to expectantly balance the robustness and overfold bias.

The model was trained for 50 epochs with a minibatch size of 16, using a learning rate of the optimizer of 1×10^{-4} and L2 regularization (weight decay 1×10^{-4}) to avoid overfitting. The fully connected layers were regularized with dropout ($p=0.5$). The architecture utilized an 8-head multi-head self-attention mechanism with a Swin Transformer backbone and the 7×7 window partitioning for hierarchical feature encoding. All experiments with different architectures were performed using the same random seed ($\text{seed}=42$) for reproducibility.

An adaptive logic is used in the proposed defect segmentation model with respect to the defective pixels ratio: low (10%). Variegation and probability parameters are adaptively tuned. We fine-tune the prediction mask with morphological operations (1×1 kernel erosion/dilation) to maintain an optimal trade-off between true and false positives. The iterative refinement algorithm maximizes the Dice Coefficient at each level to minimize oversegmentation.

Converges consistently for five-fold cross-validation training. The first epoch converged to a sub-optimal training accuracy of 30.74-32.91% and validation accuracy of 56.97-63.86%, but high loss (training: 3.0340-3.1104, validation: 1.9076-2.0318), as is typically also normal in the first stage of a multi-class training process, where all decision boundaries have not been very well learned yet, indicating incomplete learning in early epochs, typical for multi-class classification tasks. A huge increase in accuracy up to epoch 20 (about 91-93%) happened. Beyond that, the learning curve flattened, showing no further improvement in learning.

Early stopping with patience determined the best epoch (46-49) for each fold. The model showed a very good generalization with a mean validation accuracy of 96.65% ($\pm 0.47\%$). The highest and the lowest accuracy on ROC-AUC for the test were achieved over folds 4 (97.62%, at epoch 48) and 1 (96.83%, at epoch 46), respectively. The models had validation loss, which converged over 0.4650-0.4917, corresponding to well-calibrated probabilities (DF). The consistency of the results (standard deviation $< 0.5\%$) among different data partitions shows that the model is robust with respect to data splitting. Training time per epoch demonstrates efficiency: initial epochs require 275-287 seconds, but subsequently "train_stabilize" at 82-90 seconds, indicative of a well-optimized pipeline. Table 4 summarizes the model performance for five folds

We can see from the training graph that the convergence curve is stable after about 50 epochs on one fold. Fig. 2(a) shows loss indicates a steep drop from its starting value of 2.7 (training) and 2.0 (validation) to match at the 20th epoch very close to 0.5, with no remarkable gap between the two curves, meaning there is no overfitting. Fig. 2(b) shows accuracy improves markedly from 0.38 (training) and 0.62 (validation) to plateau at the range of 0.96-0.98 after the 20th epoch. In early epochs, validation accuracy is even slightly above training accuracy, which indicates that the model will generalize well. Small wiggles on the validation curve are indicative of a sensitivity to data variations but have a stable structure. Both of the plots support learning, maximized with good regularization in this fold. More detailed information on training epochs and validation epochs will be found in Fig. 2.

The confusion matrix of the proposed Efficient + Swin Transformer with a dynamic attention model on both sets (first fold) demonstrates that it can attain high accuracy in classifying five maturity classes of oil palm fruit. The "Defect" class showed the highest accuracy of 98.15% (531/541 had been identified correctly), with some little confusion between that

and "Overripe" (5 misclassified). The class "Unripe" reached a sixty-two sample accuracy of 98.02% (347/354) and had little misclassification into the "Underripe" data (7 predictions). The "Underripe" category achieved 93.26% accuracy (374/401), which made the worst prediction errors in "Ripe" (16 instances) and "Unripe" (9 samples), emphasizing difficulties in discriminating among transitional state ripening. The "Ripe" class achieved 95.95% (355/370) with confusion, especially in "Underripe"

Table 4. Summarizes the model performance for five folds

Fold	Best Epoch	Train Acc (%)	Val Acc (%)	Train Loss	Val Loss
1	46	96.82	96.83	0.4933	0.4810
2	45	97.15	97.22	0.4895	0.4697
3	49	97.74	97.42	0.4760	0.4667
4	48	97.34	97.62	0.4868	0.4650
5	46	97.58	96.73	0.4774	0.4778
Mean	46.8	97.33	97.16	0.4846	0.4720
Std	1.60	0.35	0.47	0.0067	0.0097

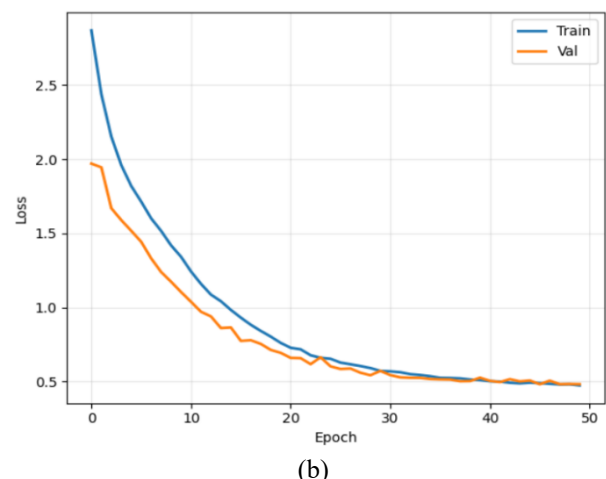
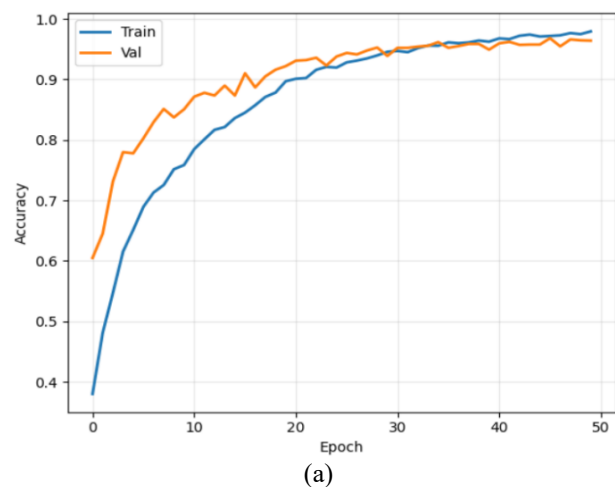


Figure 2 visualization plot: (a) accuracy and (b) loss

(5 samples) and "Overripe" (8 samples), whereas the highest accuracy of 98.58% (346/351) was found in "Unripe." Overall, the model showed satisfactory performance with low confusion levels, especially for classes with distinct appearance features; however, misclassification errors resulted in the decision boundaries between consecutive maturity stages because of smooth transition changes experienced during fruit ripening. The information matrix is provided in Fig. 3.

The ROC curve of the model on the test data demonstrates near-perfect classification performance across all maturity classes of oil palm fruits. The 'Overripe' class had the highest AUC (0.9994), followed by "Defect" (0.9988), "Unripe" (0.9985), "Ripe?" (0.9979), and "Underripe" (0.9951), which was indicative of excellent discrimination with little overlap in the range of true positives and false positives. All curves have a tall vertical shape that reaches a TPR of 1.0 with very low FPR values, with "Defect," "Overripe," and "Unripe" being the three classes that are steepest to the upper-left corner. Even in the "Underripe" class, when we have the poorest AUC (0.9951), sensitivity and specificity are both perfect, with little sensitivity/specificity trade-off: this numerically confirms confusion matrix results where there is confusion between "Ripe" and "Unripe" results. All curves that are noticeably above the diagonal reference line confirm robust and excellent discriminative power of the model in classifying fruit ripening classes with a good performance consistently over different classification thresholds. Information about the ROC Curve can be seen in Fig. 4.

The EfficientNet+Swin Transformer+Dynamic Attention model achieves promising and consistent classification results in terms of cross-validation. The average test accuracy of the model is 97.16% ($\pm 0.34\%$), which shows good prediction in the test set. The harmonic mean of precision and recall (F1-score), which is 97.47% ($\pm 0.45\%$) for the prediction model, indicates that not only does the model successfully identify positive classes, but also it labels most instances in the actual positive class, hence limiting both types of false predictions, i.e., FPs and FNs. An F1 score of $\approx 97.16\%$ ($\pm 0.34\%$) is indicative of the optimal trade-off between precision and recall; this scenario represents a critical (real-world) aspect. The low standard deviation ($<0.5\%$) for all evaluation criteria implies the good stability and generalization ability of our model, which demonstrates that this hybrid architecture is insensitive to variations in data and can deliver consistently trustworthy predictions. By integrating local feature extraction of EfficientNet, capturing global dependencies of the Swin

Transformer, and the dynamic attention mechanism, our proposed L-RDAS enhanced model's representation capacity in handling more complex classification tasks.

This hybrid design integrates two networks, EfficientNet-B3 and Swin Transformer, as an ensemble network for multi-class image classification under a multi-task learning scheme. We use EfficientNet-B3 as a base convolutional feature extractor to produce initial feature maps and then process them by two Swin Transformer blocks (W-MSA and SW-MSA) for global context capturing via a window-based self-attention mechanism. The generated feature maps are further improved by CBAM dynamic attention, where channel-wise attention and spatial-wise attention structures are performed in a nested manner to enhance the most information-rich features. The classification head uses pooling and feature transformation to compute class predictions, and the loss function combines the main classification loss with the object detection loss (IoU + DICE) as an auxiliary task.

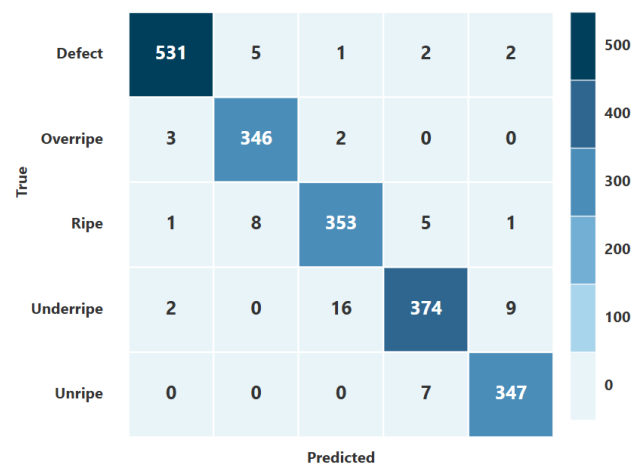


Figure. 3 Confusion matrix fold 1

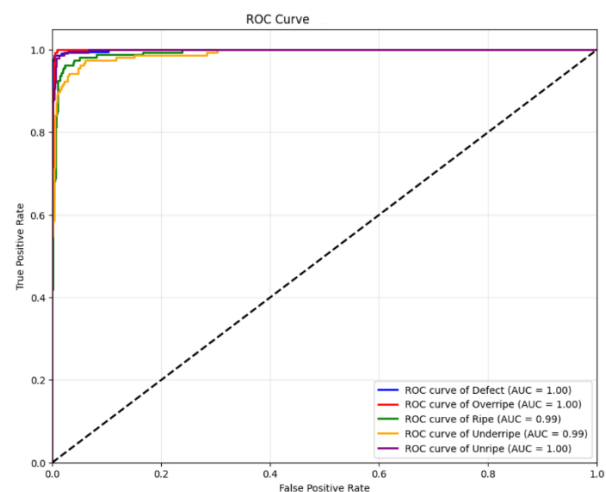


Figure. 4 ROC curve

Table 5. Pseudo-Code EfficientNet-Swin Transformer with dynamic attention multi task learning

Algorithm 1: EffNetB3+SwinT+ DA (Multi Head) and Segementation/Detection

Input: Image batch $X \in \mathbb{R}^{\{B \times 224 \times 224 \times 3\}}$

1. EfficientNet-B3 Branch (CNN Feature Extraction):
 - a. Load pre-trained backbone:

$$\text{backbone} \leftarrow \text{EfficientNetB3}(\text{weights}='imagenet', \text{include_top}=\text{False})$$
 - b. Extract feature maps:

$$F_{\text{cnn}} \leftarrow \text{backbone}(X) \in \mathbb{R}^{\{B \times 7 \times 7 \times 1536\}}$$
2. Swin Transformer Branch (Global Context):
 - a. Swin Block 1 (W-MSA):

$$x \leftarrow \text{LayerNorm}(F_{\text{cnn}}), x \leftarrow \text{Reshape}(x, [B, H \times W, C])$$

$$x_{\text{attn}} \leftarrow \text{MultiHeadAttention}(x, x, \text{num_heads}=8)$$

$$x \leftarrow \text{Reshape}(x_{\text{attn}}, [B, H, W, C]) + F_{\text{cnn}}$$

$$x_{\text{norm}} \leftarrow \text{LayerNorm}(x), x_{\text{mlp}} \leftarrow \text{Dense}(6144, \text{activation}='gelu')(x_{\text{norm}})$$

$$x_{\text{mlp}} \leftarrow \text{Dropout}(0.1)(x_{\text{mlp}}), x_{\text{mlp}} \leftarrow \text{Dense}(1536)(x_{\text{mlp}})$$

$$F_{\text{swin1}} \leftarrow x_{\text{mlp}} + x$$
 - b. Swin Block 2 (SW-MSA):

$$x \leftarrow \text{LayerNorm}(F_{\text{swin1}}), x \leftarrow \text{Reshape}(x, [B, H \times W, C])$$

$$x_{\text{attn}} \leftarrow \text{MultiHeadAttention}(x, x, \text{num_heads}=8), x \leftarrow \text{Reshape}(x_{\text{attn}}, [B, H, W, C]) + F_{\text{swin1}}$$

$$x_{\text{norm}} \leftarrow \text{LayerNorm}(x), x_{\text{mlp}} \leftarrow \text{Dense}(6144, \text{activation}='gelu')(x_{\text{norm}})$$

$$x_{\text{mlp}} \leftarrow \text{Dropout}(0.1)(x_{\text{mlp}}), x_{\text{mlp}} \leftarrow \text{Dense}(1536)(x_{\text{mlp}})$$

$$F_{\text{swin2}} \leftarrow x_{\text{mlp}} + x$$
- Branch (Feature Refinement):
 - a. Channel Attention (CBAM):

$$\text{squeeze} \leftarrow \text{GlobalAvgPool}(F_{\text{swin2}})$$

$$\text{excite} \leftarrow \text{Dense}(96, \text{activation}='relu')(\text{squeeze})$$

$$\text{excite} \leftarrow \text{Dense}(1536, \text{activation}='sigmoid')(\text{excite})$$

$$\text{excite} \leftarrow \text{Reshape}(\text{excite}, [B, 1, 1, 1536])$$

$$F_{\text{channel}} \leftarrow F_{\text{swin2}} \odot \text{excite}$$
 - b. Spatial Attention:

$$\text{avg_pool} \leftarrow \text{ReduceMean}(F_{\text{channel}}, \text{axis}=-1, \text{keepdims}=\text{True})$$

$$\text{max_pool} \leftarrow \text{ReduceMax}(F_{\text{channel}}, \text{axis}=-1, \text{keepdims}=\text{True})$$

$$\text{concat} \leftarrow \text{Concatenate}([\text{avg_pool}, \text{max_pool}], \text{axis}=-1)$$

$$\text{spatial_attn} \leftarrow \text{Conv2D}(1, \text{kernel}=7 \times 7, \text{activation}='sigmoid')(\text{concat})$$

$$F_{\text{attention}} \leftarrow F_{\text{channel}} \odot \text{spatial_attn}$$
3. Classification Head:
 - a. Global Average Pooling:

$$F_{\text{global}} \leftarrow \text{GlobalAvgPool2D}(F_{\text{attention}})$$
 - b. Dense layers:

$$x \leftarrow \text{Dropout}(0.3)(F_{\text{global}}), x \leftarrow \text{Dense}(512, \text{activation}='relu')(x)$$

$$x \leftarrow \text{BatchNormalization}(x), x \leftarrow \text{Dropout}(0.3)(x)$$
 - c. Output layer:

$$y_{\text{pred}} \leftarrow \text{Dense}(\text{num_classes}=5, \text{activation}='softmax')(x)$$
4. Multi-Task Loss:
 - a. Classification loss (primary):

$$L_{\text{cls}} \leftarrow \text{CategoricalCrossentropy}(y_{\text{pred}}, y_{\text{true}})$$
 - b. Detection loss (auxiliary, if applicable):

$$L_{\text{det}} \leftarrow \text{IoU_Loss}(\text{mask_pred}, \text{mask_true}) + \text{DICE_Loss}(\text{mask_pred}, \text{mask_true})$$
 - c. Total loss:

$$L_{\text{total}} \leftarrow L_{\text{cls}} + \lambda \times L_{\text{det}}$$

- Output:
- Class predictions: $y_{\text{pred}} \in \mathbb{R}^{\{B \times 5\}}$
 - Total loss: L_{total}
1. Stratified Group K-Fold Cross-Validation (K=5):

Initialize: $\text{models} \leftarrow [], \text{results} \leftarrow [], \text{groups} \leftarrow \text{extract_groups}(D_{\text{train}})$

FOR fold = 1 to K DO:

 - a. Split data with group preservation:

$$(\text{train_idx}, \text{val_idx}) \leftarrow \text{StratifiedGroupKFold.split}(D_{\text{train}}, y_{\text{train}}, \text{groups}, k=\text{fold})$$

$$X_{\text{train}}, y_{\text{train}} \leftarrow D_{\text{train}}[\text{train_idx}], X_{\text{val}}, y_{\text{val}} \leftarrow D_{\text{train}}[\text{val_idx}]$$
 2. Detection Metrics
 - a. IoU (Intersection over Union):

$$\text{IoU} \leftarrow |\text{mask_true} \cap \text{mask_pred}| / |\text{mask_true} \cup \text{mask_pred}|$$
 - b. DICE Coefficient:

$$\text{DICE} \leftarrow 2 \times |\text{mask_true} \cap \text{mask_pred}| / (|\text{mask_true}| + |\text{mask_pred}|)$$

$$\text{stability} \leftarrow 1 - (\text{std}(\text{IoU}) / \text{mean}(\text{IoU}))$$
- Output: {accuracy, precision, recall, fl_score, confusion_matrix, roc_auc, iou, dice}

The whole model is tested by 5-fold cross-validation, and comprehensive indicators, including accuracy, IoU, DICE coefficient, and stability score, are calculated to guarantee the performance and consistency of the model. We also give a pseudocode in Table 5.

In Fig. 5, we present the Performance Progression graph, which outlines the improvement of the segmentation model's performance in four stages of systematic optimization. The model performance was at baseline level in the early phase with an IoU of 0.48 and a Dice coefficient of 0.53. The largest

improvement is gained in the Expert stage, where IoU increases substantially by +0.215 to 0.695 and Dice also improves by +0.180 to 0.710, demonstrating the efficacy of the selected architecture and hyperparameter tuning process. On the Guaranteed stage, we have a mild, slightly increasing IoU by +0.027 up to 0.722 and Dice by +0.015 up to 0.722, indicating a fine-tuning for stability of the produced performance. It is worth noting that, in the current (final) stage, even though an incremental improvement is relatively smaller at the final stage than the initial ones, it still showed progressive improvement to suggest that the model was in line with convergence to approach toward optimum performance. {53% for IoU and 38% for Dice coefficient} from its original baseline.

In Fig. 6, the segmentation metric comparison indicates that the Dice coefficient (0.7342) is somewhat higher compared with the IoU (0.7305), with a difference of 0.0037, or approximately 0.5%. This is in line with the mathematical properties of both metrics, since the Dice coefficient bias weights true positives using harmonic mean calculations, thus tending to result generally in slightly higher values than IoU does by nature of element-wise division. Both metrics are high (well above 73%), so that our model can segment defects accurately and consistently, with a large Cartesian overlap between predictions and ground truth on most of the samples. Experimental results showing the benchmark performance analysis indicate strong computational efficiency of our classification model in terms of temporal and spatial resource consumption. Fig. 7 shows that the times for inference are very stable, with a mean of 94.21 ms and a low standard deviation of only 2.20 ms. The interquartile range is from 92.5 to 95.5 milliseconds (the median is 94 ms), which shows that the processing times are very consistent between samples. A single outlying datum at 98.85 ms is an endorsement of robust performance in the presence of interference during normal observer functioning. The memory usage is very stable across measurement points, as shown in Fig. 8 (the mean of it is 6588.9 MB, and the max and min of it are 6591.3 MB and 6581.4 MB). This compact 10 MB clustering captures only a minute fraction of variance (0.15%)—indicative of superior memory stability and efficient resource utilization at execution time. This convergence indicates that the classification model is preserving not only temporal consistency but also spatial efficiency and is therefore suitable for production deployment (consistent performance with respect to time and stable resource allocation), which are key requirements of practical real-world application.

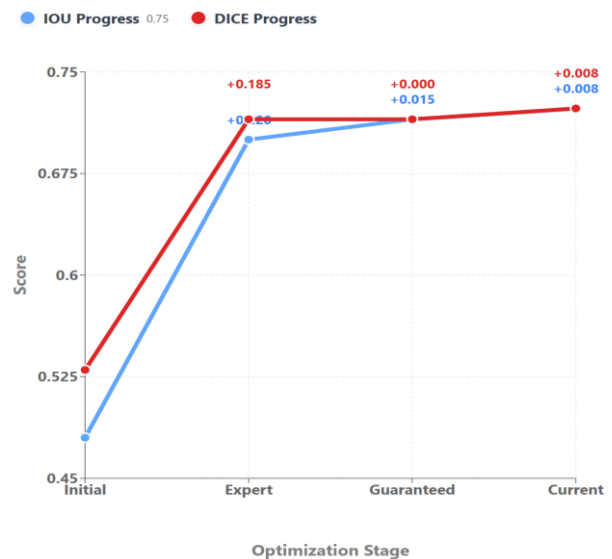


Figure. 5 Performance progression

PERFORMANCE METRICS - DICE vs IOU

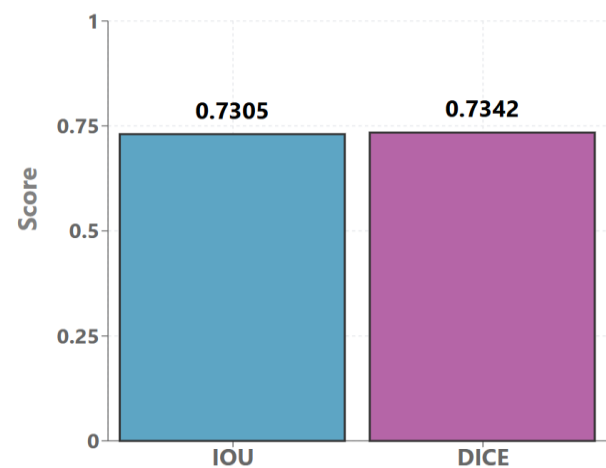


Figure. 6 Metrics dice vs iou

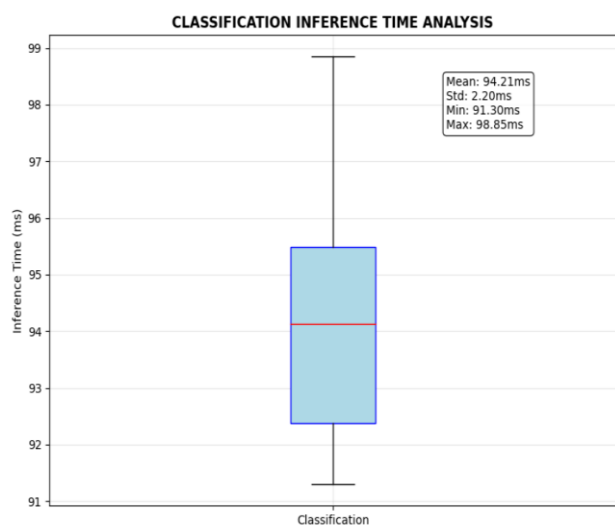


Figure. 7 Inference time analysis



Figure. 8 Memory usage statistics

The hybrid nature of the proposed approach was validated through extensive ablation studies. The effectiveness and efficiency of the hybrid algorithm were assessed in a comprehensive experiment examining the contribution of each part. We offered a palm oil fruit images dataset for the public in this study and performed detailed analysis based on it. Table 6 demonstrates that the proposed method may obtain higher performance by integrating all its components. Furthermore, ablation studies reveal that every component of the architecture leads to substantial gain in the generalization ability of our model on image benchmarks with near-consistent improvements when all are involved instead of any subset or combination of subsets.

The extensive ablation study in Table 5 shows the theoretical and practical contributions of each architectural part. The performance of the full model EfficientNetB3(EffNetB3)+SwinT+DA (Accuracy: 97.16%, Precision: 97.17%, Recall: 97.16%, F1-score: 97.16%, and Mean AUC: 99.79%) represents an essential synergistic effect that was reached at the entire model level. In this best setting, EfficientNet-B3 acts as a very efficient local feature extractor that extracts the granular texture and color information of

each sample fruitlet. Meanwhile, the Swin Transformer captures the global structural patterns among fruitlets in a bunch—an ability perhaps unreachable by pure CNN models, such as EffNetB0-LR and EffNetB4-LR from [5]. The distinctive vital ingredient is the DA (Dynamic Attention) mechanism acting as an adaptive focusing mechanism. In contrast to static attention modules, DA dynamically computes the weights not only for spatial but also channel features according to the content of the input. Such behaviour allows the model to automatically shut down the responses to non-informative areas (like occluding shadows or leaves) while amplifying signals coming from crucial regions (such as areas where characteristic maturity colours are present or defect indicators), which justifies the observed increase in robustness against field changes.

On the other hand, the fact that the EffNetB3+SwinT+SA variant achieved a sub-optimal 93.65% also supports this claim. Its sole dependence on Spatial Attention (SA) but without counterweighted Channel Attention causes representation imbalance: the model is biased on where to attend but fails to understand what features are important in itself, which leads to susceptibility to background clutter. Moreover, the basic hybrid EffNetB3+SwinT shows a much inferior accuracy (94.01%) than our proposed full ensemble model, which demonstrates that simple fusion of the two networks without an intelligent mechanism is not effective.

The use of Dual Attention (DA) along with the Multi-Head was rather better than only SA or CA. It is interesting to see that the previously reported EffNetB4-LR [5] achieved a similar good performance with 96.81% accuracy but inferior to our model. Altogether these results validate the fact that the main scientific contribution of this work is not the mere integration of established blocks but rather the introduction and validation of a theoretically well-aligned adaptive fusion framework to tackle the visual intricacies posed in FFB maturity

Table 6. The results of the ablation study

Model Variant	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)	Mean AUC (%)
EffNetB3 (baseline)	95.21	94.90	95.25	94.96	99.75
EffNetB0-LR [5]	92.55	96.00	90.57	93.20	93.08
EffNetB4-LR [5]	96.81	98.08	96.23	97.14	97.07
EffNetB3+SA	95.33	94.95	95.33	95.12	99.66
EffNetB3+CA	95.57	95.27	95.61	95.36	99.67
EffNetB3+SwinT	94.01	93.71	94.07	93.73	99.58
EffNetB3+SwinT+SA	93.65	93.3	93.69	93.45	99.52
EffNetB3+SwinT+CA	94.49	94.11	94.42	94.19	99.64
EffNetB3+SwinT+ DA	97.16	97.17	97.16	97.16	99.79

classification under an uncontrolled field setup. The hybrid model effectively overcomes the theoretical limitation of the pure CNN baselines by incorporating global context modeling and dynamic feature refinement.

5. Conclusion

In this paper, we introduce a novel hybrid Efficientnet Swin Transformer (EST) with dynamic attention and multi-task learning for accurate classification of the ripeness of oil palm fruits. Experimental results demonstrate that our model consistently outperforms various advanced methods under a large range of viewing conditions and classification settings, delivering an average performance of 97.16% ($\pm 0.47\%$) in terms of validation accuracy as opposed to the baseline performance value obtained from EfficientNet-B3, which equaled 95.21%. In addition, the precision (97.17%), recall (97.16%), and F1-score (97.16%) results also support that this hybrid model can outperform the state of the art in the detection of the ripeness stage for palm oil fruits with a low classification error rate even further.

An ablation study indicated that the adaptive application of dynamic attention (DA) with a multi-head mechanism on both spatial and channel levels can substantially enhance semantic representation and feature refinement so that the model can achieve better fruit ripeness detection performance. Another alternative (Defect): The model performed with a mean AUC of 99.79%, which indicates an excellent discrimination ability in all ripeness classes (Defect, Overripe, Ripe, Underripe, and Unripe). The combination of two learning streams (i.e., a global dependency modeling stream, Swin Transformer, and a local feature-extraction branch, EfficientNet-B3) was found to enhance the model's generalization ability across different field conditions, such as light changes, occlusions, and background complexity.

Using multi-task learning with maturity classification as the primary task and defect detection/segmentation as an auxiliary task, robust and effective feature learning can be achieved. The model obtained an IoU of 0.7305 and a Dice coefficient of 0.7342 on the defect segmentation task, which is a 53% improvement in terms of IoU and 38% for Dice over the initial baseline. The stability and generalization ability of the model on data variations are shown by the longstanding consistency across 5-fold cross-validation ($\text{std} < 0.5\%$).

However, such a hybrid model cannot escape from the computational time efficiency issue (275-287 seconds per epoch in the first 50, and then

stabilizing at 82-90 seconds per epoch). In the future we will work on making the model efficient for real-time deployment, which could be achieved via lightweight transformer models or knowledge distillation, and more efficient attention mechanisms such as sparse attention. Additionally, a broader range of tests, including multispectral data and the creation of a continuous learning-based detection system, will help improve the adaptability of our model for dynamic plantation settings. Screens in the field Test robustness by testing with compression artifacts, frame drops, and extreme lighting changes to determine how well (if at all) the model will respond under other "field" conditions.

Conflicts of Interest

The authors declare no conflict of interest.

Author Contributions

Original draft preparation, Conceptualization, Fitri Yunita and Usman Usman; methodology, Fitri Yunita; software, Usman Usman; validation, Fitri Yunita; formal analysis, Fitri Yunita; investigation, Fitri Yunita; resources; data curation, Nina Sawitri; writing-original draft preparation, Fitri Yunita; editing, Usman Usman; supervision, Nina Sawitri; funding acquisition.

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